BUILD Indonesia | Aerospace & Defense

ATC FlowLens

Explainable Runway Safety and ATC Decision-Support Dashboard

Team

Agentic Vibing

PROBLEM

Runway operations are time-critical

Controllers must connect aircraft, vehicles, runway status, weather, and communication within seconds.

Aircraft tracks

Approaching, landing, departing, or taxiing aircraft.

Vehicle movement

Airport vehicles and runway crossing requests.

Radio and readbacks

Voice instructions, blocked transmissions, and readback risk.

Unit coordination

Ground, tower, approach, airport ops, and emergency teams.

Runway status

Active runway, occupancy, crossing, takeoff, and landing state.

Weather and visibility

Fog, wind shift, storm cells, and low-visibility operations.

Flight strips

Clearance, runway, sequence, and handoff information.

EVIDENCE

Runway conflict is a real safety issue

Real incidents show how timing, communication, visibility, workload, and vehicle movement can combine into risk.

LATAM Peru 2213

Lima, 2022: aircraft collided with a fire truck on runway during takeoff roll.

LaGuardia 2026 preliminary

Reports described fire truck crossing close to landing aircraft in high workload.

Austin-Bergstrom

2023: departure aircraft was cleared while another aircraft was on final in dense fog.

Common pattern

Timing pressure, communication, vehicle movement, visibility, and workload combine.

GAP

Current systems help, but the picture can still be fragmented

ATC FlowLens is a support and training layer, not a replacement for certified systems.

Existing systems

Radar, ADS-B, MLAT, surface surveillance, electronic strips, A-SMGCS, RIMCAS, STCA, CPDLC, SWIM.

Voice remains common

Tower and ground operations still depend heavily on radio communication.

FlowLens role

Support, simulation, and training layer that makes runway risk easier to understand.

Maturity gap

Not every airport has the same surface surveillance and conflict alerting capability.

Fragmentation

Runway status, clearance, aircraft timing, and vehicle movement can sit in separate places.

SOLUTION

ATC FlowLens turns runway complexity into one operational view

A dashboard for conflict risk, workload awareness, and event replay.

Runway occupancy

Shows clear, occupied, crossing, takeoff, or inbound landing state.

Clearance state engine

Tracks cleared to land, cleared for takeoff, hold short, cross, and go around.

Explainable alert

Shows why an alert appears, not only that risk exists.

Aircraft and vehicle tracking

Displays simulated positions on a simple airport map.

Time-to-conflict

Compares aircraft touchdown or runway entry time with vehicle clear time.

Replay mode

Reviews the event timeline for training and safety analysis.

HOW IT WORKS

From simulated operational data to explainable runway risk

The MVP can prove the core logic without sensitive live ATC data.

Inputs

Aircraft position, speed, phase, runway; vehicle position, status, route; clearance, comms, weather.

Outputs

Dashboard, conflict alert, reason, recommended action category, workload indicator, replay timeline.

Processing

Runway occupancy model, clearance engine, time-to-conflict, workload score, severity classification.

Example rule

If cleared-to-land aircraft arrives before crossed vehicle clears the same runway, trigger HIGH alert.

MVP

A focused demo scenario keeps the prototype buildable

Simulation-first scope makes the idea credible for a hackathon.

BAW123

Aircraft is on final approach and cleared to land.

ARFF-2

Airport vehicle receives runway crossing clearance.

Prediction

System estimates aircraft touchdown time and vehicle runway clear time.

Conflict

Vehicle may still occupy runway when aircraft arrives.

FlowLens output

Alert, explanation, workload state, and replay timeline.

Build priority

Data model, map, clearance engine, conflict detection, alert explanation, workload, replay.

Start with training and safety awareness before live operations

A realistic adoption path creates value while avoiding premature certification claims.

Target users

Tower, ground, approach, airport ops, ARFF, ATC training centers, safety investigators.

Training value

Improves runway conflict scenario quality and post-event explanation.

Expansion path

Speech-to-text, readback mismatch detection, richer workload analytics.

Initial market

Simulation, training, and runway safety awareness for airports, aviation schools, ANSPs, and airport teams.

Operational value

Supports safety briefings, after-action review, and tower-ground-emergency coordination.

IMPACT & ALIGNMENT

Advanced technology applied to a real aerospace safety challenge

The concept connects safety, training, operational readiness, and sustainability.

Earlier risk recognition

See aircraft movement, vehicle activity, clearance status, and timing in one view.

Stronger training replay

Review why a conflict happened and what contributed.

Future sustainability

Potential to reduce avoidable go-arounds, taxi delay, holding, and fuel burn.

Better workload awareness

Expose active risk factors before task saturation.

BUILD alignment

Aerospace & Defense, Advanced Technology and Innovation, Social Impact, Sustainability.

ROADMAP

From hackathon MVP to aviation safety training platform

Phase 1

Hackathon MVP: simulated runway map, time-to-conflict alert, workload indicator, replay timeline.

Phase 2

Enhanced prototype: speech-to-text, readback mismatch, scenario generator, multi-runway support.

Phase 3

Future vision: airport data, tower-ground training simulator, safety analytics, certified support research.

ATC FlowLens helps air traffic teams see runway risk before it becomes a runway incident.

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